

3.4

Consider the z -transform $X(z)$ whose pole-zero plot is as shown in Figure P3.4.

- Determine the ROC of $X(z)$ if it is known that the Fourier transform exists. For this case, determine whether the corresponding sequence $x[n]$ is right sided, left sided, or two sided.
- How many possible two-sided sequences have the pole-zero plot shown in Figure P3.4?
- Is it possible for the pole-zero plot in Figure P3.4 to be associated with a sequence that is both stable and causal? If so, give the appropriate ROC.

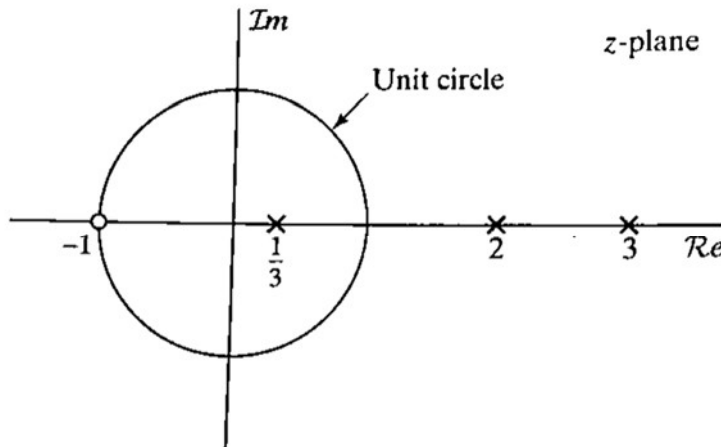


Figure P3.4

3.6

- Following are several z -transforms. For each, determine the inverse z -transform using ~~both methods—partial fraction expansion and power series expansion—discussed in Section 3.3.~~ In addition, indicate in each case whether the Fourier transform exists.

(a) $X(z) = \frac{1}{1 + \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$

(b) $X(z) = \frac{1}{1 + \frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}$

(c) $X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 + \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}, \quad |z| > \frac{1}{2}$

(d) $X(z) = \frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-2}}, \quad |z| > \frac{1}{2}$

(e) $X(z) = \frac{1 - az^{-1}}{z^{-1} - a}, \quad |z| > |1/a|$

3.9

A causal LTI system has impulse response $h[n]$, for which the z -transform is

$$H(z) = \frac{1 + z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{4}z^{-1}\right)}.$$

- (a) What is the ROC of $H(z)$?
- (b) Is the system stable? Explain.
- (c) Find the z -transform $X(z)$ of an input $x[n]$ that will produce the output

$$y[n] = -\frac{1}{3}\left(-\frac{1}{4}\right)^n u[n] - \frac{4}{3}(2)^n u[-n-1].$$

- (d) Find the impulse response $h[n]$ of the system.

3.23

An LTI system is characterized by the system function

$$H(z) = \frac{\left(1 - \frac{1}{2}z^{-2}\right)}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)}, \quad |z| > \frac{1}{2}.$$

- (a) Determine the impulse response of the system.
- (b) Determine the difference equation relating the system input $x[n]$ and the system output $y[n]$.

3.36

The pole-zero diagram in Figure P3.36 corresponds to the z -transform $X(z)$ of a causal sequence $x[n]$. Sketch the pole-zero diagram of $Y(z)$, where $y[n] = x[-n+3]$. Also, specify the ROC for $Y(z)$.

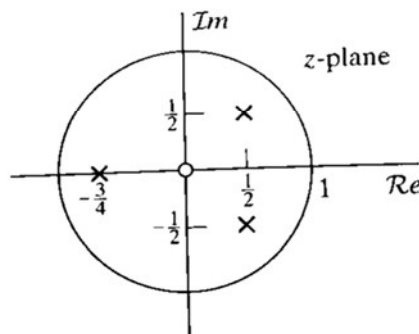


Figure P3.36

Matlab problem

(1) Find the poles and zeros of

$$G(z) = \frac{2z^4 + 16z^3 + 44z^2 + 56z + 32}{3z^4 + 3z^3 - 15z^2 + 18z - 12}$$

(2) Plot the poles and zeros on the z-plane

(3) Find the partial fraction expansion of $G(z)$